1 Computational analysis: empirical games theory analysis

Suppose we evaluate a collection of policies by simulation and find an equilibrium of the estimated game. There are two reasons it might fail in the actual interaction: the estimated payoffs may be wrong, and a profitable policy may be missing from the collection. The exercises below separate these two problems.

Research basis: Michael P. Wellman, Karl Tuyls, and Amy Greenwald (2025), *Empirical Game-Theoretic Analysis: A Survey*, Sections 6–7.

Definition 1.1 (Full and restricted games)**.** There are players $i = 1, \ldots, n$. Player i has a finite set S_i of policies. A *pure profile* $a = (a_1, \ldots, a_n) \in S := \prod_i S_i$ selects one policy per player, and $u_i(a) \in \mathbb{R}$ is player i's expected payoff from that interaction.

We have evaluated only a nonempty subset $X_i \subseteq S_i$ for each player. The *restricted game* allows policies in X_i; the *full game* allows policies in S_i. Write $X = \prod_i X_i$, and let $\hat{u}_i(a)$ be the estimated payoff for $a \in X$. The notation a_{-i} denotes the other players' policies.

Definition 1.2 (Mixed strategies and expected payoffs)**.** The set $\Delta(X_i)$ consists of probability distributions on X_i. A *mixed strategy* $\sigma_i \in \Delta(X_i)$ assigns probability $\sigma_i(a_i)$ to policy a_i. Players randomize independently. For $\sigma = (\sigma_1, \ldots, \sigma_n)$,

$$u_i(\sigma) = \sum_{a \in X} \left(\prod_j \sigma_j(a_j) \right) u_i(a),$$

$$u_i(b_i, \sigma_{-i}) = \sum_{a_{-i} \in X_{-i}} \left(\prod_{j \neq i} \sigma_j(a_j) \right) u_i(b_i, a_{-i}), \qquad X_{-i} = \prod_{j \neq i} X_j.$$

The second expression is the payoff if player i switches to b_i while everyone else keeps the same strategy. Estimated expected payoffs use the same formulas with $\hat{u}$ in place of u.

Definition 1.3 (Regret and equilibrium)**.** The *deviation gain* from switching to b_i is

$$d_i(b_i; \sigma) = u_i(b_i, \sigma_{-i}) - u_i(\sigma).$$

Restricted and full regret are the largest available deviation gains:

$$R_X(\sigma) = \max_i \max_{b_i \in X_i} d_i(b_i; \sigma), \qquad R_S(\sigma) = \max_i \max_{b_i \in S_i} d_i(b_i; \sigma).$$

Write $\widehat{R}_X$ for regret calculated from $\hat{u}$. A Nash equilibrium has regret zero; an ϵ-equilibrium has regret at most ϵ. An *empirical equilibrium* satisfies $\widehat{R}_X(\sigma) = 0$. A pure equilibrium is *strict* if every distinct unilateral deviation lowers the deviator's payoff strictly.

Checking pure deviations suffices: the payoff of a mixed deviation is an average of pure-deviation payoffs. Throughout, the candidate profile σ uses policies in X, even when deviations in the full game are considered.

Exercise 1.1 (Certifying an equilibrium from estimated payoffs).

(a) **Payoff error and regret error.** Assume $\eta \geq 0$ and

$$|u_i(a) - \hat{u}_i(a)| \leq \eta \qquad \text{for every player } i \text{ and every } a \in X.$$

(i) Prove that $|R_X(\sigma) - \widehat{R}_X(\sigma)| \leq 2\eta$ for every mixed profile σ on X.

(ii) Show that the constant 2 cannot be reduced: construct a family attaining equality for $\eta > 0$.

(iii) Explain why the bound still holds if σ is chosen after inspecting the estimated payoff table. Distinguish this deterministic statement from a claim about the probability that the payoff-error assumption holds.

(b) **Equilibrium under every admissible payoff table.** Now allow different error bounds for different entries:

$$\mathcal{U} = \{u : |u_i(a) - \hat{u}_i(a)| \leq e_i(a) \text{ for all } i, a \in X\}, \qquad e_i(a) \geq 0.$$

This set is *rectangular*: each entry may take any value in its interval, with no restriction linking it to other entries. This is a description of possible payoff tables, not a probabilistic independence assumption.

(i) Fix a pure profile $a^\star \in X$. Derive a necessary and sufficient condition for it to be a Nash equilibrium of every restricted game in $\mathcal{U}$.

(ii) Give the corresponding condition for strict equilibrium. Explain what happens at equality.

(iii) Suppose the payoff errors must move together. Identify which direction of your proof can fail, and give an example.

(c) **Policies missing from the table.** Assume the error bound in Exercise 1.1(a). A deviation-search procedure also certifies

$$\max_{s_i \in S_i} u_i(s_i, \sigma_{-i}) \leq \max_{a_i \in X_i} u_i(a_i, \sigma_{-i}) + \delta_i, \qquad \delta_i \geq 0.$$

The guarantee concerns all policies in S_i, including those the procedure did not evaluate.

(i) Derive an upper bound on $R_S(\sigma)$ in terms of $\widehat{R}_X(\sigma)$, η, and the δ_i.

(ii) Suppose some $X_i \subsetneq S_i$ and omitted-policy payoffs have no known bound. Prove that the restricted payoff table alone gives no finite upper bound on full-game regret. Keep the restricted table fixed in your construction.

(iii) Explain what evidence would justify the displayed search guarantee. Is failure to find a profitable deviation sufficient?

(d) **Selecting an equilibrium from noisy estimates.** Suppose that, for each fixed entry separately,

$$\Pr\{|u_i(a) - \hat{u}_i(a)| \le \eta\} \ge 1 - \alpha, \qquad 0 < \alpha < 1.$$

The true table is fixed; the estimates are random. This is *marginal coverage*. *Simultaneous coverage* would bound the probability that all required inequalities hold together. A selection rule is *adaptive* if it chooses σ using the observed estimates.

(i) Does marginal coverage imply $\Pr\{R_X(\sigma) \le 2\eta\} \ge 1 - \alpha$ for every adaptively selected empirical equilibrium? Prove this or give a counterexample, specifying both the estimator and the selection rule.

(ii) Describe a valid certificate based on simultaneous coverage. State exactly which errors must be controlled.

(iii) Describe a second certificate using independent data collected after the candidate is fixed. State what must be checked in those data.

Exercise 1.2 (A fixed mixed equilibrium under payoff uncertainty).

Use $\mathcal{U}$ from Exercise 1.1(b), with every $e_i(a) > 0$. The *support* of σ_i is the set of actions assigned positive probability. A profile is *genuinely mixed* if some player has at least two actions in their support.

(a) Can one fixed genuinely mixed profile be an exact equilibrium for every $u \in \mathcal{U}$? Prove your answer. The same profile must work throughout the uncertainty set.

(b) Explain why a strict pure equilibrium can behave differently. Distinguish robustness of one fixed profile from existence of some equilibrium in each game.

2 Normative agency: the separation principle

An agent repeatedly decides whether to bet on a coin whose type is hidden. We compare two rules: declining can either end the task permanently or skip just one bet. The coin and the information available to the agent are otherwise the same.

This question is adapted from Massachusetts Institute of Technology: Dimitri P. Bertsekas, 6.231: Dynamic Programming and Stochastic Control, **Fall 2008 midterm, Problem 2**, hosted in the Fall 2015 archive. Questions, pp. 3–4; solutions, pp. 4–5.

The coin and the rewards. There are N stages, $k = 0, \dots, N-1$. The hidden type X_k is either h or $\bar{h}$, where $0 < \bar{h} < h < 1$. Conditional on X_k, the next toss is a head with probability X_k. After the toss, the type switches to the other type with probability $q \in [0, 1]$. Switches are independent over time, and tosses are conditionally independent given the type sequence. All parameters and the initial prior $p_0 = \Pr(X_0 = h)$ are known.

The agent acts before seeing the toss. Betting earns $r > 0$ on a head and $-\ell < 0$ on a tail; declining earns zero. The objective is expected total reward over the remaining stages, without discounting or wealth constraints. Actions do not affect the coin process.

Rule	What declining does
Stopping	Ends play permanently. All later rewards are zero.
Resumption	Skips the current bet. Its outcome is still observed, and later bets remain available.

In either problem, betting reveals the outcome. The order is therefore: choose an action, observe the toss and receive the reward, then let the hidden type switch.

Definition 2.1 (Belief and belief update)**.** Before toss k, the agent's belief is

$$p_k = \Pr(X_k = h \mid z_0, \dots, z_{k-1}), \qquad z_j \in \{H, T\}.$$

Two probabilities must be distinguished:

$$p = \Pr(\text{high type}), \qquad \theta(p) = ph + (1-p)\bar{h} = \Pr(\text{head on the next toss}).$$

The expected reward of betting is $g(p) = r\theta(p) - \ell[1 - \theta(p)]$. After observing z, Bayes' rule gives the posterior about the current type:

$$B_H(p) = \frac{ph}{\theta(p)}, \qquad B_T(p) = \frac{p(1-h)}{1-\theta(p)}.$$

After the possible switch, the belief for the next stage is

$$p_{k+1} = \Phi(p, z) := q + (1-2q)B_z(p).$$

These update formulas may be used in the exercises.

Definition 2.2 (Policies, information states, and values)**.** An admissible policy chooses an action using the stage and observed history, possibly with independent private randomization. It cannot use the hidden type or future tosses.

An information state is *sufficient* if histories with the same state have the same available actions and, conditional on each action, the same distribution of the next reward and information state.

Let $V_k(p)$ be the optimal expected remaining reward under stopping, and $W_k(p)$ the corresponding value under resumption, starting active at stage k with belief p. The Bellman principle compares each action's immediate expected reward plus its expected future value. A *myopic* rule compares immediate rewards only.

For any function F of the next belief, write its expected value after observing a toss as

$$(\mathcal{K}F)(p) := \theta(p)F(\Phi(p,H)) + [1-\theta(p)]F(\Phi(p,T)).$$

This notation is just a weighted average of the head and tail branches.

Exercise 2.1 (Information, future choices, and optimal behavior).

(a) **The state and the Bellman equations.**

(i) Prove that (k, p_k) is a sufficient state while play is active. Specify what must be added to represent permanent stopping.

(ii) Derive the recursions for V_k and W_k, with terminal conditions and the value of the stopped state. Write the value of each available action before taking a maximum.

(iii) Prove $W_k(p) \geq V_k(p)$ by showing how a resumption policy can reproduce any stopping policy.

(iv) Explain why (k, p_k) may cease to be sufficient if parameters are unknown or preferences depend on accumulated wealth.

(b) **When is the myopic rule optimal?**

(i) Use the resumption recursion to determine the optimal action when $g(p) > 0$, $g(p) < 0$, and $g(p) = 0$.

(ii) Now let the next-belief distributions depend on the action: $K_B(\cdot \mid p)$ for betting and $K_S(\cdot \mid p)$ for skipping. Here $K_a(A \mid p)$ is the probability of a next belief in a measurable set $A \subseteq [0,1]$. The expected continuation value is $\int F(p')K_a(dp' \mid p)$. Characterize exactly when

$$\int F\, dK_B = \int F\, dK_S \quad \text{for every bounded measurable } F.$$

A bounded measurable function has a finite bound on its absolute value and is measurable for the usual Borel sets. Explain why equality for one particular value function is a weaker condition.

(iii) If only betting reveals the outcome, must the optimal current action change? Give a precise answer, including a case in which it does not.

(c) **Betting now to preserve a later choice.** Consider two remaining stages, relabeled 0 and 1. Let $q = 0$, so the hidden type is fixed. Assume

$$0 < \bar{h} < t < h < 1, \qquad t = \frac{\ell}{r + \ell}.$$

Thus t is the head probability at which one bet breaks even. Define $[x]_+ = \max\{0, x\}$ and

$$C(p) = \theta(p)[g(B_H(p))]_+ + [1 - \theta(p)][g(B_T(p))]_+.$$

This is the expected value of the final opportunity after the first toss is observed.

(i) Express $V_0(p)$ and $W_0(p)$ using $g(p)$ and $C(p)$.

(ii) Prove that, for every allowed parameter tuple, a nonempty open interval of beliefs satisfies $g(p) < 0 < g(p) + C(p)$. A useful starting point is the belief $p^\star$ satisfying $\theta(p^\star) = t$.

(iii) On that interval, determine the uniquely optimal current action in each problem and calculate $W_0(p) - V_0(p)$. Explain why the actions differ.

(d) **What can be inferred from optimal behavior?** Two controllers are behaviorally equivalent if they select the same actions, or action distributions, at every feasible observation history.

(i) Construct a controller that reproduces an optimal policy for this fixed finite task without computing a posterior at decision time. What does this show about inferring an internal algorithm from behavior?

(ii) Let s_k include the stage, belief, and stopped status, and let $r_k(s_k, a_k, s_{k+1})$ be the reward on a transition. Include the observed outcome in the next state if needed to represent the realized reward. For bounded state functions Ψ_k, consider

$$r'_k = c\, r_k + \Psi_{k+1}(s_{k+1}) - \Psi_k(s_k), \qquad c > 0, \quad \Psi_N \equiv 0.$$

Keep the transitions and admissible policies fixed. Prove that r' and r give the same ranking of policies from a fixed initial state. Treat stopped play as absorption with zero original rewards through the common horizon, and transform those transitions too.

(iii) Explain why ending the calculation at a policy-dependent stopping time can invalidate that argument. State what further restrictions or evidence would be needed to identify rewards or implementation.

3 Descriptive explanations: the Hugging Face incident

A worker may hold back output because extra output reduces coworkers' pay, or because exceeding a cutoff is punished. We first ask whether those explanations can predict the same choice. We then ask what an intervention would have to establish to distinguish them.

This question is adapted from Massachusetts Institute of Technology: Frank Schilbach, 14.13: Psychology and Economics, Problem Set 4, Part 1, Questions 5 and 9–10. Questions, pp. 1–2; solutions, pp. 1–4. The course archive is **Spring 2020**; the PDFs' colophons say **Spring 2021**.

The pay mechanism. There are $N \geq 2$ workers. Worker i chooses output $x_i \geq 0$, receives pay y_i, and incurs production cost $c(x_i)$:

$$y_i = px_i - \frac{\beta}{N-1}\sum_{j\neq i} x_j, \qquad v_i = y_i - c(x_i), \qquad p, \beta > 0.$$

Here p is pay per unit of own output, β measures the reduction due to coworkers' average output, and v_i is material payoff after production costs. The same formula applies to every worker, so increasing one's own output reduces each coworker's pay.

Assume c is twice continuously differentiable, with

$$c'(0) = 0, \qquad c''(x) > 0, \qquad \lim_{x\to\infty} c'(x) = \infty.$$

Thus marginal cost c' is strictly increasing from zero and eventually exceeds any fixed reward rate. Throughout a best-response calculation, hold coworkers' outputs fixed.

Definition 3.1 (The two objectives)**.** *Model A* assigns positive weight α to each coworker's material payoff:

$$U_i^A = v_i + \alpha\sum_{j\neq i} v_j, \qquad \alpha > 0, \quad p > \alpha\beta.$$

Model C cares only about own material payoff and an imposed penalty:

$$U_i^C = v_i - D\mathbf{1}\{x_i > x_c\}, \qquad D \geq 0, \quad x_c \geq 0.$$

The indicator is one above the cutoff and zero at or below it. In particular, $x_i = x_c$ is not penalized. A best response maximizes the specified objective over $x_i \geq 0$; it need not be unique. The penalty is treated as enforceable for now, without modeling the incentives of whoever imposes it.

Exercise 3.1 (Distinguishing motives from behavior).

(a) **Two explanations for the same output.** Let x_A maximize Model A, and let x_S maximize own material payoff v_i alone. Two models are *observationally equivalent for a given observation* if both can produce that observation.

(i) Derive x_A and x_S. Prove that each exists and is unique, and determine their ordering.

(ii) Set $x_c = x_A$. Find the necessary and sufficient condition on D for x_A to maximize Model C. Describe all optimal choices when the condition holds with equality.

(iii) Express the material gain from exceeding the cutoff as an integral. Explain why the comparison does not require quadratic costs, and what it leaves unresolved about credible enforcement.

(b) **An intervention that could distinguish the models.** An intervention changes a condition of the decision problem. An *exclusion assumption* specifies the channel through which it affects choice; a *stability assumption* specifies which other features remain fixed.

(i) Design an intervention on monitoring or enforceability that gives different predictions under Models A and C while leaving the pay mechanism fixed. State both predictions.

(ii) State the exclusion and stability assumptions required to interpret an observed change as evidence about the mechanism. Does random assignment alone establish them?

(iii) Relax one assumption and construct another explanation for the same observed change in output.

(c) **Inferring altruism when motives can coexist.** Now drop $p > \alpha\beta$ and allow both motives:

$$U_i^z = v_i + \alpha \sum_{j \neq i} v_j - D_z \mathbf{1}\{x_i > x_c\}, \qquad \alpha \geq 0.$$

The index z denotes the intervention condition. Suppose condition $z = 0$ credibly removes punishment, $D_0 = 0$, with known p, β, c unchanged. Let x_0 be the individual's optimal output in this condition.

The *identified set* consists of all values of α compatible with an observation and the assumptions. A singleton gives point identification; multiple compatible values give partial identification; an empty set contradicts the model.

(i) Derive the no-punishment choice $x_0(\alpha)$, including the boundary $x_0 = 0$.

(ii) For each possible observed output, determine the identified set for α. Include outputs inconsistent with the assumptions.

(iii) Suppose workers have different α_i and only mean output $\bar{x} = N^{-1} \sum_i x_{i,0}$ is observed. Does this identify mean altruism $\bar{\alpha} = N^{-1} \sum_i \alpha_i$? Explain separately the effects of nonlinear marginal costs and zero-output choices, and give conditions under which the mean does suffice.

(d) **Applying the same inference problem to artificial agents.** A finite nonempty set of contexts Z records one action $a^\star(z)$ in each context. The common finite action set A contains at least three actions. A reward function $R : Z \times A \to \mathbb{R}$ *rationalizes* these choices if

$$a^\star(z) \in \arg\max_{a \in A} R(z, a) \quad \text{for every } z \in Z.$$

The maximizer may be tied. A positive affine transformation has the form $R' = \lambda R + b$ with common constants $\lambda > 0$ and $b \in \mathbb{R}$.

(i) Construct two reward functions that rationalize every observed choice but are not positive affine transformations of one another.

(ii) Specify a restricted reward class and interventions that could distinguish altruism from enforcement. State which features and parameters must remain fixed across contexts.

(iii) Explain how observations could falsify that class. Distinguish this from uniquely identifying its parameters and from establishing what objective the agent internally represents.

Solutions

Computational analysis: empirical games

Solution to Exercise 1.1.

Exercise 1.1(a): Payoff error and regret error

(i) Prove the bound.

We need to control two operations: averaging payoffs and comparing the largest deviation gains.

First, averaging cannot increase a uniform error bound. If $w(a)$ is the probability of pure profile a under any mixed profile τ, then $w(a) \geq 0$ and $\sum_a w(a) = 1$. Hence

$$\begin{aligned} |u_i(\tau) - \hat{u}_i(\tau)| &= \left| \sum_{a \in X} w(a)[u_i(a) - \hat{u}_i(a)] \right| \\ &\leq \sum_{a \in X} w(a)|u_i(a) - \hat{u}_i(a)| \leq \eta. \end{aligned}$$

This applies to both $\tau = \sigma$ and $\tau = (b_i, \sigma_{-i})$.

A deviation gain subtracts these two expected payoffs. Define

$$\hat{d}_i(b_i; \sigma) = \hat{u}_i(b_i, \sigma_{-i}) - \hat{u}_i(\sigma).$$

The triangle inequality gives

$$\begin{aligned} |d_i(b_i; \sigma) - \hat{d}_i(b_i; \sigma)| &\leq |u_i(b_i, \sigma_{-i}) - \hat{u}_i(b_i, \sigma_{-i})| \\ &\quad + |u_i(\sigma) - \hat{u}_i(\sigma)| \leq 2\eta. \end{aligned}$$

Thus every true deviation gain is at most the corresponding estimated gain plus 2η. Taking maxima gives $R_X \leq \widehat{R}_X + 2\eta$. Interchanging true and estimated payoffs gives $\widehat{R}_X \leq R_X + 2\eta$. Together,

$$\boxed{|R_X(\sigma) - \widehat{R}_X(\sigma)| \leq 2\eta.}$$

In particular, an empirical equilibrium has true restricted regret at most 2η.

(ii) Show that the constant is sharp.

For any $\eta > 0$, take one player with two actions:

	A	B
estimated payoff $\hat{u}$	0	0
true payoff u	$-\eta$	η

Each entry has error η. At the pure choice A, estimated regret is zero, but switching to B improves the true payoff by 2η. Therefore no smaller universal constant works. Extra players with only one available action turn this into a multiplayer example without changing the calculation.

(iii) Allow the profile to depend on the estimates.

Let

$$E = \{|u_i(a) - \hat{u}_i(a)| \leq \eta \text{ for every } i, a \in X\}.$$

For any realized table in E, the proof in (i) holds for *every* σ. It therefore holds for a profile selected from that table. No independence between selection and estimation is required.

This establishes the implication $E \Rightarrow |R_X - \widehat{R}_X| \leq 2\eta$. It does not establish $\Pr(E)$. That probability requires a simultaneous guarantee over the payoff entries, as discussed in Exercise 1.1(d).

Exercise 1.1(b): Equilibrium under every admissible table

(i) Derive and prove the condition.

For any entry, define its lower and upper endpoints by

$$L_i(a) = \hat{u}_i(a) - e_i(a), \qquad U_i(a) = \hat{u}_i(a) + e_i(a).$$

The condition is

$$\boxed{L_i(a^\star) \geq U_i(b_i, a^\star_{-i}) \quad \text{for every } i \text{ and } b_i \neq a^\star_i.}$$

Equivalently, the estimated payoff advantage must cover both error radii:

$$\hat{u}_i(a^\star) - \hat{u}_i(b_i, a^\star_{-i}) \geq e_i(a^\star) + e_i(b_i, a^\star_{-i}).$$

Sufficiency. If the endpoint condition holds, then every admissible table satisfies

$$u_i(a^\star) \geq L_i(a^\star) \geq U_i(b_i, a^\star_{-i}) \geq u_i(b_i, a^\star_{-i}).$$

No unilateral deviation is profitable, so $a^\star$ is an equilibrium.

Necessity. If the condition fails for some deviation, set the incumbent payoff to its lower endpoint and the deviating payoff to its upper endpoint. Keep all other entries at their estimated values. These are distinct entries, and rectangularity permits both choices at once. The resulting admissible game has a profitable deviation. Thus $a^\star$ is not an equilibrium of every admissible game.

(ii) Distinguish strict equilibrium and equality.

Strict equilibrium in every admissible game requires

$$L_i(a^\star) > U_i(b_i, a^\star_{-i}) \quad \text{for every distinct deviation.}$$

If equality holds for a deviation, the endpoint construction makes the player indifferent. The profile remains an equilibrium but is not strict in that game.

(iii) Explain what changes under coupled errors.

Sufficiency survives if the actual uncertainty set is a subset of the displayed intervals. Necessity may fail because opposite endpoints may no longer be jointly attainable.

For example, suppose every payoff of player i receives the same unknown shift z_i:

$$u_i(a) = \hat{u}_i(a) + z_i \qquad \text{for every } a.$$

Then the shift cancels from every payoff comparison. An equilibrium of the estimated game remains an equilibrium for every allowed shift, even if the separate intervals overlap. The proof of necessity above fails at the step that chooses the two entries independently.

Exercise 1.1(c): Policies missing from the table

(i) Combine the two error bounds.

For player i, let

$$R_{X,i} = \max_{a_i \in X_i} u_i(a_i, \sigma_{-i}) - u_i(\sigma),$$

and define $R_{S,i}$ and $\widehat{R}_{X,i}$ analogously. Subtracting the same incumbent payoff from the search guarantee gives

$$R_{S,i} \le R_{X,i} + \delta_i.$$

The argument of Exercise 1.1(a) also applies to each player's maximum, so $R_{X,i} \le \widehat{R}_{X,i} + 2\eta$. Therefore

$$\begin{aligned} R_S(\sigma) &\le \max_i \{\widehat{R}_{X,i}(\sigma) + 2\eta + \delta_i\} \\ &\le \widehat{R}_X(\sigma) + 2\eta + \max_i \delta_i. \end{aligned}$$

The last bound has three terms: regret in the estimated game, the effect of payoff error, and the possible gain from omitted policies. At an empirical equilibrium the first term is zero.

(ii) Prove that the restricted table alone is insufficient.

Choose a player with an omitted policy $t_i \in S_i \setminus X_i$. Fix the true table on X and the candidate σ. Its payoff $u_i(\sigma)$ is now fixed.

For any $M > 0$, extend the table by assigning

$$u_i(t_i, a_{-i}) = u_i(\sigma) + M \quad \text{for every } a_{-i} \in X_{-i}.$$

These entries were not evaluated. Averaging them against σ_{-i} gives

$$u_i(t_i, \sigma_{-i}) - u_i(\sigma) = M,$$

so $R_S(\sigma) \ge M$. Because M can be arbitrarily large while the restricted table remains unchanged, no finite bound follows from that table alone.

Both qualifications matter. If every $X_i = S_i$, no policy is omitted. If all payoffs lie in a known interval $[L, U]$, then $U - L$ is a finite upper bound, though possibly uninformative.

(iii) Explain what makes the search guarantee valid.

Not finding a deviation only describes the policies examined. A bound over *all* policies needs, for example, exhaustive search, a proved approximation guarantee, or valid upper bounds on unexplored policies. Errors in the search procedure's payoff estimates must also be controlled, since the displayed guarantee concerns true payoffs.

If the payoff and search guarantees fail with probabilities at most α and β, respectively, both hold with probability at least $1 - \alpha - \beta$ by the union bound. Independence is unnecessary.

Exercise 1.1(d): Selecting an equilibrium from noisy estimates

(i) Give a counterexample.

The proposed implication is false. Consider one player with actions $a_0, a_1, \ldots, a_m$, where $m \geq 2$. Choose $B > 2\eta$. The baseline action is known exactly:

$$u(a_0) = \hat{u}(a_0) = 0.$$

Each other action has true payoff $-B$. Independently across $j \geq 1$, estimate it by

$$\hat{u}(a_j) = \begin{cases} +B, & \text{with probability } \alpha, \\ -B, & \text{with probability } 1-\alpha. \end{cases}$$

A fixed nonbaseline entry is exact with probability $1-\alpha$ and has error $2B > \eta$ otherwise. Every entry therefore has the required marginal coverage.

Select an action maximizing the estimated payoff. If any error occurs, an inferior action appears strictly better than a_0, so the rule selects an inferior action. Its true regret is $B > 2\eta$. If no error occurs, it selects a_0 and has regret zero. Consequently,

$$\Pr\{R_X(\sigma) > 2\eta\} = 1 - (1-\alpha)^m > \alpha.$$

The selected action is always an empirical equilibrium. The failure arises because selection can exploit an error in *any* of the m entries, whereas the assumption controls each entry separately. Dummy players embed the example in a multiplayer game.

(ii) Certify all relevant entries simultaneously.

Let M be the number of payoff entries in the restricted game. Construct estimates for which each entry exceeds its error tolerance η with probability at most α/M. The union bound then gives

$$\Pr\{\text{some entry has error greater than } \eta\} \leq M\frac{\alpha}{M} = \alpha.$$

With probability at least $1-\alpha$, the bound in Exercise 1.1(a) holds for every profile, including an adaptively selected one. Existing marginal $1-\alpha$ intervals cannot simply be relabeled as simultaneous intervals; their calibration or the amount of data must change. If sampling stops adaptively, the error guarantee must also be valid under that stopping rule.

(iii) Fix the candidate and validate it with fresh data.

After selecting σ, collect independent validation data. Conditional on the selection data, σ is fixed. Construct a *joint* event of probability at least $1-\alpha$ on which

$$\begin{aligned} |u_i(\sigma) - \tilde{u}_i(\sigma)| &\leq b_i, \\ |u_i(a_i, \sigma_{-i}) - \tilde{u}_i(a_i, \sigma_{-i})| &\leq c_{i,a_i} \quad \text{for every } i, a_i \in X_i. \end{aligned}$$

On that event, subtracting the lower bound on the incumbent payoff from the upper bound on each deviation gives

$$R_X(\sigma) \leq \max_{i,a_i} \{\tilde{u}_i(a_i, \sigma_{-i}) - \tilde{u}_i(\sigma) + c_{i,a_i} + b_i\}.$$

This can certify small regret or reveal that the candidate fails validation. Estimating only its incumbent payoff is insufficient: regret compares that payoff with all relevant deviations. Neither remedy controls omitted policies; that still requires Exercise 1.1(c).

Solution to Exercise 1.2.

(a) A fixed genuinely mixed profile cannot be robust here.

The center table $\hat{u}$ belongs to $\mathcal{U}$. If σ is not an equilibrium there, it already fails the requirement. Otherwise choose a player i who assigns positive probability to at least two actions, and choose one supported action a_i.

At equilibrium, every supported action has the same maximal expected payoff against σ_{-i}. Otherwise the player could move probability from a worse action to a better one. In particular,

$$\hat{u}_i(a_i, \sigma_{-i}) = \hat{u}_i(\sigma), \qquad 0 < \sigma_i(a_i) < 1.$$

Choose an opponent profile a_{-i} with positive probability w under σ_{-i}. Increase just the entry $u_i(a_i, a_{-i})$ by d, where $0 < d \leq e_i(a_i, a_{-i})$, leaving other entries at their centers. This is allowed because the radius is positive and the set is rectangular.

The pure action a_i gains expected payoff dw. The mixed strategy gains only $\sigma_i(a_i)dw$, since it plays a_i only that fraction of the time. The new gain from deviating to a_i is therefore

$$(1 - \sigma_i(a_i))dw > 0.$$

Thus the same mixed profile is not an equilibrium of every admissible game.

(b) Strict pure equilibria can have a protective margin.

A strict pure equilibrium uses strict payoff inequalities, which can survive small perturbations. The endpoint condition in Exercise 1.1(b) states exactly how large a margin is needed for the given uncertainty set.

A mixed equilibrium instead requires exact indifference among supported actions. The construction breaks that equality. It does not rule out a different mixed equilibrium in the perturbed game; its mixing probabilities may change.

Normative agency: the separation principle

Solution to Exercise 2.1.

Exercise 2.1(a): The state and the Bellman equations

(i) Show that the belief summarizes the relevant history.

Consider two active histories at the same stage k with the same belief p. In both histories,

$$\Pr(z_k = H) = \theta(p), \qquad \Pr(z_k = T) = 1 - \theta(p).$$

After outcome z, Bayes' rule gives $B_z(p)$ for the current type. The switch then gives the next belief

$$\Phi(p, z) = (1 - q)B_z(p) + q[1 - B_z(p)].$$

Thus the two histories have the same distribution of the next observation and belief. Each available action also gives the same immediate reward distribution and the same future action restrictions. No earlier feature of the history changes these quantities once (k, p) is known.

At stage N, both histories have zero remaining value. Backward induction now shows that they have the same optimal continuation values at every earlier stage. This proves sufficiency while active. To cover permanently stopped histories, add an active/stopped indicator: a stopped history has no further betting choices regardless of its belief.

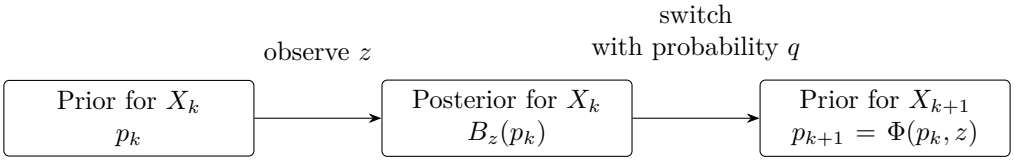


Figure 1: Filtering followed by prediction. In the resumption problem this same update occurs after either betting or skipping, because both actions reveal the outcome and leave the hidden process unchanged.

(ii) Derive both recursions.

Recall that

$$(\mathcal{K}F)(p) = \theta(p)F(\Phi(p, H)) + [1 - \theta(p)]F(\Phi(p, T)).$$

This is the expected next-stage value if the next-stage value function is F. The action values are

Problem	Decline	Bet
Stopping	0	$g(p) + (\mathcal{K}V_{k+1})(p)$
Resumption	$(\mathcal{K}W_{k+1})(p)$	$g(p) + (\mathcal{K}W_{k+1})(p)$

In stopping, declining removes every future reward. In resumption, skipping still reveals the toss and preserves later choices. Maximizing the two entries in each row gives

$$V_k(p) = \max\{0,\ g(p) + (\mathcal{K}V_{k+1})(p)\},$$
$$W_k(p) = \max\{(\mathcal{K}W_{k+1})(p),\ g(p) + (\mathcal{K}W_{k+1})(p)\}.$$

The terminal conditions are $V_N(p) = W_N(p) = 0$. A stopped state is absorbing and has reward and remaining value zero at every stage.

(iii) Compare the sets of available policies.

Take any stopping policy. In the resumption problem, make the same bets until that policy stops; from then on, skip every bet. Ignore the extra observations obtained after stopping would have occurred.

This resumption policy earns exactly the same total reward on each coupled sequence of coin outcomes. Resumption can therefore reproduce every stopping policy. Maximizing over its larger set of policies gives

$$\boxed{W_k(p) \geq V_k(p).}$$

The inequality compares optimal values, not necessarily the actions chosen at the current stage.

(iv) Identify the assumptions behind sufficiency.

If parameters such as h, $\bar{h}$, or q are unknown, histories with the same $p = \Pr(X_k = h)$ can imply different beliefs about those parameters. Their future predictions may differ. One generally needs the joint posterior over the hidden state and unknown parameters.

If wealth affects utility or which bets are feasible, histories with the same (k, p) can leave different wealth levels. Wealth must then be included in the state. In each case, the proposed state omits information relevant to future decisions.

Exercise 2.1(b): When is the myopic rule optimal?

(i) Compare betting with skipping.

In the base resumption model, both actions have the same continuation term. Their difference is

$$[g(p) + (\mathcal{K}W_{k+1})(p)] - (\mathcal{K}W_{k+1})(p) = g(p).$$

Hence the optimal action is

$$\boxed{\begin{cases} \text{bet}, & g(p) > 0, \\ \text{skip}, & g(p) < 0, \\ \text{either action}, & g(p) = 0. \end{cases}}$$

Equivalently, compare $\theta(p)$ with $\ell/(r + \ell)$. Future rewards matter to the value, but they cancel from this action comparison because both actions leave the same future opportunities and information.

(ii) Characterize cancellation for every continuation function.

At a fixed current belief p,

$$\int F\, dK_B = \int F\, dK_S \text{ for every bounded measurable } F \quad \Longleftrightarrow \quad K_B(\cdot \mid p) = K_S(\cdot \mid p).$$

For the forward implication, choose $F = \mathbf{1}_A$ for any measurable set A. Its expectation is the probability of A, so the assumed equality gives $K_B(A \mid p) = K_S(A \mid p)$ for every A. The two probability measures are therefore equal. For the reverse implication, equal measures give equal integrals by definition.

Equality for just one value function does not identify the measures. For example, every probability measure gives the same expectation to a constant function, even when the measures differ.

(iii) Explain when information changes the action.

If betting alone reveals the toss, betting and skipping generally induce different next-belief distributions. Their action-value difference becomes

$$Q_B(p) - Q_S(p) = g(p) + \int W_{k+1}(p')K_B(dp' \mid p) - \int W_{k+1}(p')K_S(dp' \mid p).$$

The optimal action depends on the sign of this *total* difference.

Different information need not change that sign. At the final stage, $W_N = 0$, so both integrals vanish and the myopic rule is still exact. Earlier, the information may not affect any later choice, or its value may be too small to outweigh the immediate reward difference. A different posterior alone is therefore insufficient to establish a different optimal action.

Exercise 2.1(c): Betting now to preserve a later choice

(i) Write the two values with two stages left.

At the final stage, both problems take a bet exactly when its expected reward is positive. Their final-stage value is $[g(p)]_+$. Since $q = 0$, the next belief after outcome z is $B_z(p)$. The expected value of the final opportunity is therefore the given function

$$C(p) = \theta(p)[g(B_H(p))]_+ + [1 - \theta(p)][g(B_T(p))]_+.$$

Stopping now gives zero, whereas betting gives $g(p) + C(p)$. Under resumption, skipping gives $C(p)$ and betting gives $g(p) + C(p)$. Thus

$$\boxed{V_0(p) = \max\{0, g(p) + C(p)\}, \qquad W_0(p) = C(p) + \max\{0, g(p)\}.}$$

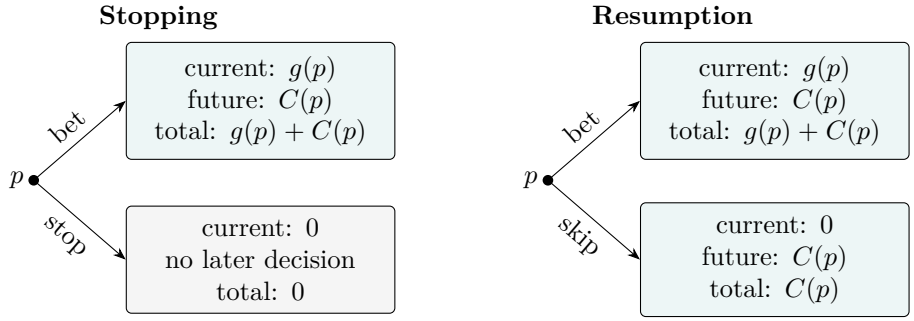


Figure 2: Expected rewards with two stages left. The head and tail observations, and the optimal final choice after each, are averaged into $C(p)$. Only the stopping branch removes that opportunity. If $g(p) < 0 < g(p) + C(p)$, the left tree selects bet and the right tree selects skip.

(ii) Find an interval where the current bet loses but continuing is worthwhile.

We will prove $g(p) < 0 < g(p) + C(p)$ just below the break-even belief. Solving $\theta(p^\star) = t$ gives

$$p^\star = \frac{t - \bar{h}}{h - \bar{h}} \in (0, 1), \qquad g(p) = (r + \ell)(h - \bar{h})(p - p^\star).$$

Therefore $g(p^\star) = 0$ and $g(p) < 0$ whenever $p < p^\star$.

At $p^\star$, a head raises the posterior above break-even and a tail lowers it:

$$B_H(p^\star) - p^\star = \frac{p^\star(h-t)}{t} > 0,$$
$$B_T(p^\star) - p^\star = \frac{p^\star(t-h)}{1-t} < 0.$$

Because g is strictly increasing, only the head branch contributes positive final-stage value. Substitution yields

$$\begin{aligned} C(p^\star) &= t\, g(B_H(p^\star)) \\ &= t(r+\ell)(h-\bar{h})[B_H(p^\star) - p^\star] \\ &= (r+\ell)(h-\bar{h})p^\star(h-t) \\ &= (r+\ell)(t-\bar{h})(h-t) > 0. \end{aligned}$$

This is the essential strict inequality: at current break-even, observing a head creates a profitable later opportunity, while after a tail the agent can decline.

Both posterior maps are continuous on $[0,1]$, because $\theta(p)$ and $1-\theta(p)$ never vanish there. The functions g, $[\,\cdot\,]_+$, and C are consequently continuous. Since

$$g(p^\star) + C(p^\star) = C(p^\star) > 0,$$

there is an $\varepsilon > 0$, with $p^\star - \varepsilon > 0$, such that $g(p) + C(p) > 0$ throughout $(p^\star - \varepsilon, p^\star)$. On the same interval $g(p) < 0$. This proves the requested open interval for every parameter tuple satisfying $\bar{h} < t < h$.

(iii) Read off the actions and the value gap.

On that interval:

- Under stopping, betting gives $g + C > 0$, while stopping gives zero. Betting is uniquely optimal.
- Under resumption, skipping gives C, while betting gives $g + C < C$. Skipping is uniquely optimal.

Consequently,

$$\boxed{W_0(p) - V_0(p) = C(p) - [g(p) + C(p)] = -g(p) > 0.}$$

With permanent stopping, the unfavorable current bet is the price of keeping the final opportunity. Resumption preserves that opportunity and the same information while avoiding the current loss.

Exercise 2.1(d): What can be inferred from optimal behavior?

(i) Construct a different implementation of the same policy.

The horizon and observation alphabet are finite, so there are finitely many feasible observation histories. Before execution, store an optimal action for each history, resolving ties consistently. At decision time, look up the current history and return its stored action.

This controller reproduces the optimal policy without calculating a posterior or solving a Bellman equation online. A randomized policy can similarly be stored as an action distribution at each history. The table may have been computed using Bayesian reasoning beforehand, but the observed actions do not establish that such reasoning occurs at decision time. Matching this fixed task is enough to refute the claim that behavior uniquely identifies the implementation.

(ii) Prove that the transformed reward preserves rankings.

Fix the initial state s_0. On any complete trajectory, the potential differences telescope:

$$\begin{aligned}\sum_{k=0}^{N-1} r'_k &= c\sum_{k=0}^{N-1} r_k + \sum_{k=0}^{N-1}[\Psi_{k+1}(s_{k+1}) - \Psi_k(s_k)] \\ &= c\sum_{k=0}^{N-1} r_k + \Psi_N(s_N) - \Psi_0(s_0) \\ &= c\sum_{k=0}^{N-1} r_k - \Psi_0(s_0).\end{aligned}$$

Every intermediate potential occurs once with a plus sign and once with a minus sign. The terminal term is zero by assumption.

Let $J_\pi(s_0)$ and $J'_\pi(s_0)$ be expected total reward under policy π before and after the transformation. Taking expectations gives

$$J'_\pi(s_0) = cJ_\pi(s_0) - \Psi_0(s_0).$$

For any two policies $\pi, \tilde{\pi}$,

$$J'_\pi(s_0) - J'_{\tilde{\pi}}(s_0) = c[J_\pi(s_0) - J_{\tilde{\pi}}(s_0)].$$

Since $c > 0$, the difference has the same sign under both rewards. Every strict ranking and every tie is preserved. Bounded potentials and a finite horizon ensure these expectations are finite.

(iii) Explain the role of the horizon and the limits of identification.

After stopping, include the dummy transitions through stage N, with zero original rewards. Their transformed rewards complete the telescoping sum above.

If instead the sum ends at a policy-dependent time τ, the remaining potential term is $\Psi_\tau(s_\tau) - \Psi_0(s_0)$. Its expectation can depend on the policy, so rankings need not be preserved without an additional terminal-potential condition.

To identify rewards, one must restrict the admissible reward class, specify which transformations count as equivalent, and obtain observations that distinguish the remaining candidates. To identify an implementation, one needs evidence about internal computation, such as interventions on that computation. Optimal actions alone establish neither conclusion.

Descriptive explanations: the Hugging Face incident

Solution to Exercise 3.1.

Exercise 3.1(a): Two explanations for the same output

(i) Derive the two preferred outputs.

Hold coworkers' outputs fixed and write $x = x_i$. Terms that do not depend on x cannot change the maximizing choice. The varying part of own material payoff is

$$f(x) = px - c(x).$$

Increasing x by one unit reduces each coworker's pay by $\beta/(N-1)$. Across the $N-1$ coworkers, the total marginal effect is $-\beta$. Model A's varying objective is therefore

$$f_A(x) = (p - \alpha\beta)x - c(x).$$

Its derivative is $p - \alpha\beta - c'(x)$, whereas the own-payoff objective has derivative $p - c'(x)$.

The function c' is continuous, strictly increasing from zero, and unbounded. Each of the positive numbers $p - \alpha\beta$ and p is therefore attained exactly once. The corresponding choices are

$$\boxed{x_A = (c')^{-1}(p - \alpha\beta), \qquad x_S = (c')^{-1}(p).}$$

In each case the objective's derivative is positive below the root and negative above it, proving that the root is the unique global maximum. Since $0 < p - \alpha\beta < p$ and $(c')^{-1}$ is increasing,

$$\boxed{0 < x_A < x_S.}$$

Altruism reduces output here because extra output lowers coworkers' relative pay.

(ii) Find the penalty that reproduces x_A.

Set $x_c = x_A$. Model C chooses x to maximize

$$f(x) - D\mathbf{1}\{x > x_A\}.$$

Separate the choice set into two regions:

- For $0 \le x \le x_A$, no penalty is charged. Since f increases up to $x_S > x_A$, the best choice in this region is x_A.
- For $x > x_A$, the same penalty D is charged at every output. The best choice in this region is therefore the unconstrained material optimum x_S.

It remains to compare $f(x_A)$ with $f(x_S) - D$. Let

$$\Delta = f(x_S) - f(x_A) > 0.$$

The advantage of choosing x_A is $D - \Delta$. Thus the full set of optimal choices is

$$\boxed{\begin{cases} \{x_A\}, & D > \Delta, \\ \{x_A, x_S\}, & D = \Delta, \\ \{x_S\}, & D < \Delta. \end{cases}}$$

In particular, x_A is optimal exactly when $D \ge \Delta$. At equality it is not the unique prediction: x_S is equally good.

(iii) Express the gain as an integral and interpret the result.

By the fundamental theorem of calculus,

$$\Delta = f(x_S) - f(x_A) = \int_{x_A}^{x_S} [p - c'(x)]\, dx.$$

The integrand is positive below x_S and zero at x_S. Hence the integral is positive. This derivation uses the increasing marginal cost and the ordering $x_A < x_S$, with no quadratic-cost assumption.

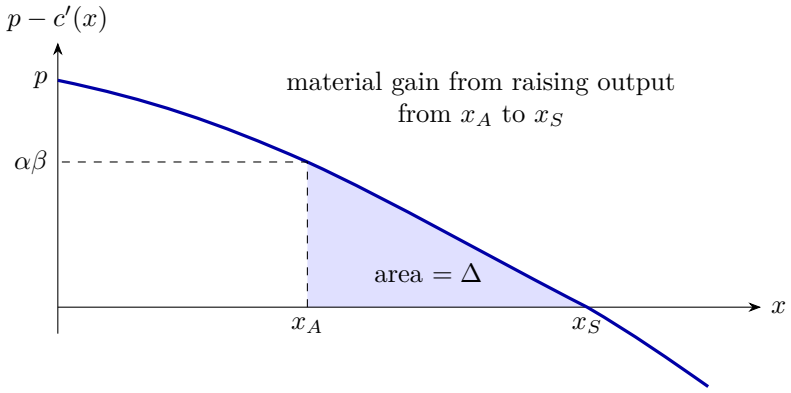


Figure 3: The required penalty offsets the entire shaded gain, not just the marginal incentive at the cutoff. Since $c'' > 0$, the curve $p - c'(x)$ is decreasing; its value at x_A is $\alpha\beta$ and at x_S is zero. The curve is schematic; the area identity holds for every cost function satisfying the assumptions.

For $D > \Delta$, both models uniquely select x_A. Observing this output alone therefore cannot distinguish them. The calculation assumes the penalty is imposed. It says nothing about who imposes it, what doing so costs, or whether punishment is credible when the time comes. Those questions require a model of the enforcer's choices and incentives.

Exercise 3.1(b): An intervention that could distinguish the models

(i) State the intervention and its predictions.

Compare a regime with enforceable punishment $D_1 > \Delta$ to one that removes punishment entirely, $D_0 = 0$. Keep the pay mechanism, cost function, and cutoff fixed. The specified models predict

Model	Punishment available	Punishment removed
Model A: fixed $\alpha > 0$	x_A	x_A
Model C: enforcement only	x_A	x_S

Model A does not contain a punishment term, so removing punishment changes none of its incentives. Model C becomes the own-material-payoff problem and selects $x_S > x_A$.

(ii) State what is required to interpret the comparison.

The intended inference requires that:

- punishment is actually unavailable and the worker understands this; disabling one monitor is insufficient if another source of punishment remains;
- p, β, the cost function, and the worker's preference parameters remain fixed;
- the intervention does not also change information, social expectations, or other future consequences relevant to choice. Coworker responses must be controlled or included in the model.

These assumptions specify both the intended channel and the features held fixed. With suitable compliance and interference assumptions, random assignment can identify the causal effect of assigning the regime. It does not by itself show which channel produced that effect.

(iii) Reproduce the effect through another mechanism.

Set punishment to zero in both regimes. Instead, let the intervention change the weight placed on coworkers' welfare:

$$\alpha_1 = \alpha > 0 \quad \text{when monitored,} \qquad \alpha_0 = 0 \quad \text{when unmonitored.}$$

Then the outputs are

$$x_1 = (c')^{-1}(p - \alpha\beta) = x_A, \qquad x_0 = (c')^{-1}(p) = x_S.$$

This is exactly the pattern predicted by enforcement, although no punishment occurs. For example, monitoring might alter the salience of coworkers' welfare. This construction violates preference stability. Randomizing monitoring does not rule it out: the intervention could causally affect that channel too.

The comparison can reject the fixed-preference altruism-only model under its assumptions. It cannot, from the treatment effect alone, rule out every account involving altruistic preferences.

Exercise 3.1(c): Inferring altruism when motives can coexist

(i) Derive the no-punishment choice, including zero output.

When $D_0 = 0$, the varying objective is

$$(p - \alpha\beta)x - c(x), \qquad x \geq 0.$$

If $p - \alpha\beta > 0$, its derivative starts positive and crosses zero once. If $p - \alpha\beta \leq 0$, its derivative is strictly negative for every $x > 0$, so zero is the unique optimum. Consequently,

$$\boxed{x_0(\alpha) = \begin{cases} (c')^{-1}(p - \alpha\beta), & 0 \leq \alpha < p/\beta, \\ 0, & \alpha \geq p/\beta. \end{cases}}$$

At $\alpha = p/\beta$, the derivative at zero is zero, but the derivative at every positive x is negative. Equality therefore belongs in the zero-output case.

(ii) Infer the compatible values of α.

For a positive choice, the first-order condition can be inverted:

$$\alpha = \frac{p - c'(x_0)}{\beta}.$$

Write $x_S = (c')^{-1}(p)$. The complete answer is

Observation	Compatible values of α
$0 < x_0 < x_S$	$\{[p - c'(x_0)]/\beta\}$
$x_0 = x_S$	$\{0\}$
$x_0 = 0$	$[p/\beta, \infty)$
$x_0 > x_S$	None: would require $\alpha < 0$
$x_0 < 0$	None: output is constrained to be nonnegative

The interior formula must not be applied at zero. Every $\alpha \geq p/\beta$ produces the same zero output, so the observation gives a lower bound rather than a unique value. An inconsistent observation rejects the joint assumptions; it does not identify which assumption failed.

(iii) Determine what mean output reveals.

First suppose every worker chooses positive output. Each satisfies

$$\alpha_i = \frac{p - c'(x_{i,0})}{\beta}.$$

Averaging gives

$$\bar{\alpha} = \frac{p - N^{-1} \sum_i c'(x_{i,0})}{\beta}.$$

Thus the required statistic is mean *marginal cost*. Mean output alone generally does not determine it.

Here is a finite-population counterexample. Choose $u, v \in (0, x_S)$ and let $m = (u + v)/2$. In population A, all N workers produce m. In population B, two workers produce u and v, while the others still produce m. Both populations have mean output m. Assign each worker the altruism parameter implied by the interior first-order condition; all those parameters are positive. Their mean altruism values differ by

$$\bar{\alpha}_A - \bar{\alpha}_B = \frac{c'(u) + c'(v) - 2c'(m)}{N\beta}.$$

If continuous c' is not affine on the relevant interval, some u, v make this expression nonzero. Otherwise midpoint equality everywhere would force c' to be affine. This proves that the same mean output can be compatible with different mean altruism even when all choices are interior.

There is an important sufficient condition. If $c'(x) = a + \kappa x$ on the known output range and all choices are interior, then

$$\boxed{\bar{\alpha} = \frac{p - a - \kappa \bar{x}}{\beta}.}$$

Mean output then suffices. For quadratic costs $c(x) = \kappa x^2/2$ up to a constant, $a = 0$.

Zero-output workers create a separate problem. Raising their α_i anywhere above p/β leaves their outputs unchanged but raises mean altruism. This ambiguity persists even with quadratic costs and even if all individual outputs are observed. Recovering mean altruism then needs further restrictions on the parameters of those workers.

Exercise 3.1(d): Applying the inference problem to artificial agents

(i) Construct different rewards with the same choices.

Give the observed action $a^\star(z)$ reward zero in every context, and give every unselected action a negative reward. The observed action is then the unique maximizer.

Pick one context and two unselected actions b, d; these exist because there are at least three actions. For positive λ, μ, ν with $\mu \neq \nu$, define

	$a^\star(z)$	b	d
R	0	$-\lambda$	$-\mu$
$\widetilde{R}$	0	$-\lambda$	$-\nu$

Keep the two assignments identical in every other entry, with all unselected rewards negative. Both rewards uniquely rationalize every recorded choice.

Suppose nevertheless that $\widetilde{R} = \kappa R + b_0$ for constants $\kappa > 0, b_0$. The common zero reward forces $b_0 = 0$. The common reward $-\lambda$ forces $\kappa = 1$. The last column would then require $\mu = \nu$, a contradiction. The rewards are not positive affine transformations of one another.

More observations in a finite set of contexts do not eliminate this freedom if arbitrary context-dependent rewards remain admissible.

(ii) Restrict the rewards and choose informative interventions.

For example, consider

$$U_{\alpha,\gamma}(a, z) = m(a, z) + \alpha w(a, z) - \gamma P(a, z), \qquad \alpha, \gamma \geq 0,$$

where m is own material return, w is others' welfare, and P is a penalty. Assume all three features are known. Normalize the coefficient on m to one and require the same α, γ across contexts. These restrictions prevent fitting an unrelated reward to each observed choice.

Use two kinds of intervention:

- With penalties disabled, change welfare consequences while controlling material incentives.
- Change enforceability while holding material incentives and welfare consequences fixed.

The interventions must be large enough to make the competing accounts predict different optimal actions somewhere. Altruism only is the restriction $\gamma = 0$; enforcement only is $\alpha = 0$. Interpreting the results requires the specified feature changes and stable coefficients.

(iii) Turn observations into a falsifiable condition.

For every context and every alternative a, the observed choice must satisfy

$$\begin{aligned} 0 \leq\ & m(a^\star, z) - m(a, z) \\ & + \alpha[w(a^\star, z) - w(a, z)] - \gamma[P(a^\star, z) - P(a, z)]. \end{aligned}$$

Intersect all these inequalities with $\alpha, \gamma \geq 0$.

- If the intersection is empty, no parameter pair in the class explains all observations. The class is falsified under the maintained assumptions.
- If exactly one pair remains, the parameters are identified within that class and normalization.
- If several pairs remain, the observations only partially identify them.

Imposing $\gamma = 0$ or $\alpha = 0$ before taking the intersection tests the corresponding single-motive account. Weak inequalities allow ties in the observed choices.

Even a unique fitted pair does not establish that the agent internally represents or computes this objective. That is an additional claim about implementation, requiring additional evidence.